Geon Lee

geonlee0325@kaist.ac.kr | geonlee0325.github.io | Seoul, South Korea

RESEARCH SUMMARY

My research goal is to uncover the mechanisms by which complex interaction structures, across relational data, retrieval systems, and LLM-based and multi-agent systems, shape learning and emergent behavior, and to develop principled frameworks for designing reliable, generalizable, and robust AI systems.

WORK EXPERIENCE

KAIST — Seoul, South Korea
Research Fellow (Advisor: Prof. Kijung Shin) — Mar. 2026 – Present

Snap Research — Bellevue, WA, USA
Research Intern (Mentors: Liam Collins, Clark Ju, Bhuvesh Kumar, Tong Zhao | Manager: Neil Shah) — Apr. – Jun. 2025
Data Augmentation for Generative Recommendation

NEC Labs America — Princeton, NJ, USA
Research Intern (Mentor: Wenchao Yu | Manager: Haifeng Chen) — May – Aug. 2023
Multi-Modal Time Series Forecasting

Amazon — San Francisco, CA, USA
Applied Scientist Intern (Mentor: Zhonghao Luo | Manager: Tao Ye) — Sep. – Dec. 2022
Conversational Music Recommendation for Alexa AI

EDUCATION

Korea Advanced Institute of Science and Technology (KAIST) — Seoul, South Korea
M.S. & Ph.D. in Artificial Intelligence — Sep. 2020 – Feb. 2026
Advisor: Prof. Kijung Shin
PhD Dissertation Award, KAIST College of Engineering

Sungkyunkwan University (SKKU) — Suwon, South Korea
B.S. in Computer Science and Engineering — Mar. 2016 – Aug. 2019
GPA: 4.41/4.50; Major GPA: 4.45/4.50
Ranked 1st in the College of CSE

PUBLICATIONS

[1] Toward a Science of AI Agent Societies
Geon Lee*, Fanchen Bu*, Soo Yong Lee*, Sunwoo Kim, and Kijung Shin
KDD 2026 (Blue Sky)

[2] A Tutorial on Retrieval-Augmented Generation for LLM-Based Recommender Systems
Sunwoo Kim*, Geon Lee*, Kyungho Kim*, Liam Collins, Neil Shah, and Kijung Shin
KDD 2026 (Tutorial)

[3] Rethinking Contrastive Learning for Graph Collaborative Filtering: Limitations and A Simple Remedy
Geon Lee, Sunwoo Kim, Kyungho Kim, and Kijung Shin
ICML 2026

[4] Hybrid-Vector Retrieval for Visually Rich Documents: Combining Single-Vector Efficiency and Multi-Vector Accuracy
Juyeon Kim, Geon Lee, Dongwon Choi, Taeuk Kim, and Kijung Shin
ACL 2026 (Findings)

[5] ItemRAG: Item-Based Retrieval-Augmented Generation for LLM-Based Recommendation
Sunwoo Kim, Geon Lee, Kyungho Kim, Jaemin Yoo, and Kijung Shin
SIGIR 2026 (Short Paper)

[6] ReFuGe: Feature Generation for Prediction Tasks on Relational Databases with LLM Agents
Kyungho Kim, Geon Lee, Juyeon Kim, Dongwon Choi, Shinhwan Kang, and Kijung Shin
WWW 2026 (Short Paper)

[7] Sequential Data Augmentation for Generative Recommendation
Geon Lee, Bhuvesh Kumar, Clark Mingxuan Ju, Tong Zhao, Kijung Shin, Neil Shah, and Liam Collins
WSDM 2026

[8] A Comprehensive Survey on Multi-Behavior Recommender Systems: Extended Taxonomy and Recent Advances
Kyungho Kim, Sunwoo Kim, Geon Lee, Jinhong Jung, and Kijung Shin
International Journal of Data Science and Analytics (2026)

[9] RDB2G-Bench: A Comprehensive Benchmark for Automatic Graph Modeling of Relational Databases
Dongwon Choi, Sunwoo Kim, Juyeon Kim, Kyungho Kim, Geon Lee, Shinhwan Kang,
Myunghwan Kim, and Kijung Shin
NeurIPS 2025 (Dataset & Benchmark)

[10] TimeXL: Explainable Multi-modal Time Series Prediction with LLM-in-the-Loop
Yushan Jiang, Wenchao Yu, Geon Lee, Dongjin Song, Kijung Shin, Wei Cheng, Yanchi Liu, and Haifeng Chen
NeurIPS 2025

[11] Attributed Hypergraph Generation with Realistic Interplay Between Structure and Attributes
Jaewan Chun, Seokbum Yoon, Minyoung Choe, Geon Lee, and Kijung Shin
ICDM 2025 *Best Paper Award; Best-Ranked Paper*

[12] Identifying Group Anchors in Real-World Group Interactions Under Label Scarcity
Fanchen Bu, Geon Lee, Minyoung Choe, and Kijung Shin
ICDM 2025

[13] A Self-Supervised Mixture-of-Experts Framework for Multi-behavior Recommendation
Kyungho Kim, Sunwoo Kim, Geon Lee, and Kijung Shin
CIKM 2025

[14] Revisiting LightGCN: Unexpected Inflexibility, Inconsistency, and A Remedy Towards Improved Recommendation
Geon Lee, Kyungho Kim, Fanchen Bu, Langzhang Liang, and Kijung Shin
Transactions on Recommender Systems (Accepted, 2025)

[15] SkySearch: Satellite Video Search at Scale
Minyoung Choe*, Geon Lee*, Changhun Han*, Suji Kim, Woong Hu, Hyebeen Hwang, Geunseok Park,
Byeongyeon Kim, Hyesook Lee, Ha-Myung Park, and Kijung Shin
KDD 2025 (Applied Data Science)

[16] KGMEL: Knowledge Graph-Enhanced Multimodal Entity Linking
Juyeon Kim, Geon Lee, Taeuk Kim, and Kijung Shin
SIGIR 2025 (Short Paper)

[17] MARIOH: Multiplicity-Aware Hypergraph Reconstruction
Kyuhan Lee, Geon Lee, and Kijung Shin
ICDE 2025

[18] A Survey on Hypergraph Mining: Patterns, Tools, and Generators
Geon Lee*, Fanchen Bu*, Tina Eliassi-Rad, and Kijung Shin
ACM Computing Surveys (SCI, 2025)

[19] Multi-Behavior Recommender Systems: A Survey
Kyungho Kim, Sunwoo Kim, Geon Lee, Jinhong Jung, and Kijung Shin
PAKDD 2025 (Survey) *Best Survey Paper Award*

[20] Beyond Neighbors: Distance-Generalized Graphlets for Enhanced Graph Characterization
Yeongho Kim, Yuyeong Kim, Geon Lee, and Kijung Shin
WWW 2025

[21] TimeCAP: Learning to Contextualize, Augment, and Predict Time Series Events with Large Language Model Agents
Geon Lee, Wenchao Yu, Kijung Shin, Wei Cheng, and Haifeng Chen
AAAI 2025

[22] Resource2Box: Learning to Rank Resources in Distributed Search Using Box Embedding
Ulugbek Ergashev, Geon Lee, Kijung Shin, Eduard Dragut, and Weiyi Meng
ICDM 2024

[23] Revisiting LightGCN: Unexpected Inflexibility, Inconsistency, and A Remedy Towards Improved Recommendation
Geon Lee, Kyungho Kim, and Kijung Shin
RecSys 2024 (Short Paper) *Best Short Paper Candidate*

[24] Post-Training Embedding Enhancement for Long-Tail Recommendation
Geon Lee, Kyungho Kim, and Kijung Shin
CIKM 2024 (Short Paper)

[25] Towards Better Utilization of Multiple Views for Bundle Recommendation
Kyungho Kim, Sunwoo Kim, Geon Lee, and Kijung Shin
CIKM 2024 (Short Paper)

[26] Representative and Back-in-Time Sampling from Real-World Hypergraphs
Minyoung Choe, Jaemin Yoo, Geon Lee, Woonsung Baek, U Kang, and Kijung Shin
Transactions on Knowledge Discovery from Data (SCIE, 2024)

[27] VilLain: Self-Supervised Learning on Homogeneous Hypergraphs without Features via Virtual Label Propagation
Geon Lee, Soo Yong Lee, and Kijung Shin
WWW 2024

[28] Mining of Real-World Hypergraphs: Patterns, Tools, and Generators
Geon Lee, Jaemin Yoo, and Kijung Shin
KDD 2023 & **WWW 2023** (Tutorial)

[29] Random Walk with Restart on Hypergraphs: Fast Computation and an Application to Anomaly Detection
Jaewan Chun, Geon Lee, Kijung Shin, and Jinhong Jung
Data Mining and Knowledge Discovery (SCI, 2023)

[30] Hypergraph Motifs and Their Extensions Beyond Binary
Geon Lee*, Seokbum Yoon*, Jihoon Ko, Hyunju Kim, and Kijung Shin
The VLDB Journal (SCI, 2023) *KAIST Outstanding Paper Award*

[31] Hypercore Decomposition for Non-Fragile Hyperedges: Concepts, Algorithms, Observations, and Applications
Fanchen Bu, Geon Lee, and Kijung Shin
Data Mining and Knowledge Discovery (SCI, 2023)

[32] Temporal Hypergraph Motifs
Geon Lee and Kijung Shin
Knowledge and Information Systems (SCIE, 2023)

[33] Set2Box: Similarity Preserving Representation Learning for Sets
Geon Lee, Chanyoung Park, and Kijung Shin
ICDM 2022

[34] HashNWalk: Hash and Random Walk Based Anomaly Detection in Hyperedge Streams
Geon Lee, Minyoung Choe, and Kijung Shin
IJCAI 2022

[35] MiDaS: Representative Sampling from Real-World Hypergraphs
Minyoung Choe, Jaemin Yoo, Geon Lee, Woonsung Baek, U Kang, and Kijung Shin
WWW 2022

[36] Mining of Real-World Hypergraphs: Patterns, Tools, and Generators
Geon Lee, Jaemin Yoo, and Kijung Shin
ICDM 2022 & **CIKM 2022** (Tutorial)

[37] Simple Epidemic Models with Segmentation Can Be Better than Complex Ones
Geon Lee, Se-eun Yoon, and Kijung Shin
PLOS ONE (SCIE, 2022) *Oral, epiDAMIK @ KDD 2021*

[38] THyMe+: Temporal Hypergraph Motifs and Fast Algorithms for Exact Counting
Geon Lee and Kijung Shin
ICDM 2021 *Best-Ranked Paper*

[39] How Do Hyperedges Overlap in Real-World Hypergraphs? — Patterns, Measures, and Generators
Geon Lee*, Minyoung Choe*, and Kijung Shin
WWW 2021

[40] Hypergraph Motifs: Concepts, Algorithms, and Discoveries
Geon Lee, Jihoon Ko, and Kijung Shin
VLDB 2020

[41] MEGA: Multi-View Semi-Supervised Clustering of Hypergraphs
Joyce Jiyoung Whang, Rundong Du, Sangwon Jung, Geon Lee, Barry Drake, Qingqing Liu, Seonggoo Kang, and Haesun Park
VLDB 2020

[42] Hyperlink Classification via Structured Graph Embedding
Geon Lee, Seonggoo Kang, and Joyce Jiyoung Whang
SIGIR 2019 (Short Paper)

ACADEMIC SERVICE

Conference Reviewer

- ACM CIKM 2022 – 2026
- ACM SIGKDD (KDD) 2023 – 2026
- NeurIPS 2024 – 2026
- AAAI 2024 – 2026
- The Web Conference (WWW) 2024 – 2026
- ACM RecSys 2025 – 2026
- ICLR 2025 – 2026
- AISTATS 2025 – 2026
- ICML 2025 – 2026
- ACM MM 2025 – 2026
- Learning on Graphs (LoG) 2022 – 2025
- ACML 2025

- SDM 2024

Journal Reviewer

- IEEE TKDE 2023 – 2026
- Information Sciences 2024 – 2026
- IEEE TPAMI 2025 – 2026
- Information Processing & Management 2025 – 2026
- Scientific Reports 2025 – 2026
- Expert Systems With Applications 2026
- Journal of Neuroscience Methods 2026
- Neurocomputing 2026
- Knowledge-Based Systems 2026
- IEEE TNNLS 2023 – 2025
- The VLDB Journal 2023 – 2025
- PLOS ONE 2024 – 2025
- Data Mining and Knowledge Discovery 2024 – 2025
- ACM TOIS 2025
- Neural Networks 2025
- Machine Learning 2025
- Patterns 2025
- IEEE TNSE 2024
- Big Data Research 2024

Session Chair

- ACM CIKM 2024

AWARDS & HONORS

Gold Reviewer, ICML 2026 May 2026
PhD Dissertation Award, KAIST College of Engineering Feb. 2026
Best Paper Award, ICDM 2025 Dec. 2025
One of the Best-Ranked Papers, ICDM 2025 Dec. 2025
Best Survey Paper Award, PAKDD 2025 Jun. 2025
Outstanding Reviewer, KDD 2025 Dec. 2024
KAIST Outstanding Paper Award Nov. 2024
Best Short Paper Candidate, RecSys 2024 Oct. 2024
One of the Best-Ranked Papers, ICDM 2021 Dec. 2021
Sungkyunkwan Presidential Award Aug. 2019
Sungkyunkwan Software Scholarship (Full tuition, all semesters) 2016 – 2019

TEACHING

Teaching Assistant

- KAIST AI506: Data Mining and Search Spring 2021, 2023
- KAIST AI607: Graph Mining and Social Network Analysis Fall 2020–2023
- KAIST AI617: Machine Learning for Robotics Spring 2022
- SKKU CSE3036: Seminar in Computer Engineering Fall 2019

Tutorial Tutor

- Mining of Real-world Hypergraphs: Patterns, Tools, and Generators CIKM 2022, ICDM 2022, WWW 2023, KDD 2023
- A Tutorial on Retrieval-Augmented Generation for LLM-Based Recommender Systems KDD 2026

Geust Lecture

- Recent Advances in Recommender Systems (KAIST AI506) May 2026

TALKS

- Understanding Complex Interaction Systems with AI: From Human Networks to AI Agent Societies — Yonsei University, Mar. 2026